

Lab 9. Simulating IoT Telemetry with MQTT

Tasks

- Did you notice a delay between clicking "Publish" and seeing the message in the "Subscriber" window? Why is low latency important for a warehouse fire sensor?

Answer: I noticed a tiny delay, but it is usually almost instant. That delay is just the time it takes for the message to travel from my publisher to the broker, and then from the broker to my subscriber. Low latency means the data travels very fast. This is incredibly important for a warehouse fire sensor because fast, real-time data delivery is essential to trigger alarms immediately and help people evacuate before a fire spreads.

- Did the "Device" need to know anything about the "Subscriber" (IP address, OS, etc.) to send the data? Explain how the Broker facilitates this "anonymity."

Answer: No, the device does not need to know anything about the subscriber, such as its IP address or operating system, because the two are completely independent of each other. The broker facilitates this anonymity by acting as a secure middleman, similar to a post office, the device simply publishes its data to a specific topic on the broker without ever knowing who is listening, while the subscriber only needs to subscribe to that same topic to receive the data. Because the broker handles the entire delivery process behind the scenes, the sender and receiver never have to interact directly, which keeps their identities and technical details private and allows the system to function without either side needing to know who the other is.

- What happens to the "System" if the "Device" sends a message to the wrong topic (e.g., warehouse/sensor/temp instead of your specific name)?

Answer: Answer: If the "Device" sends a message to the wrong topic such as warehouse/sensor/temp instead of my specific warehouse/sensor/paula/temp, the "System" will not receive the message at all. Because MQTT is topic-based, the broker only delivers messages to clients that are explicitly subscribed to that exact topic string. Since my monitoring system is only "listening" to my unique channel, it will ignore any messages published elsewhere, effectively treating the "wrong" message as if it does not exist for my specific project.

